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# Intermittent demand: Linking forecasting to inventory obsolescence

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ABSTRACT

The standard method to forecast intermittent demand is that by Croston. This method is available in ERP-type solutions such as SAP and specialised forecasting software packages (e.g. Forecast Pro), and often applied in practice. It uses exponential smoothing to separately update the estimated demand size and demand interval whenever a positive demand occurs, and their ratio provides the forecast of demand per period. The Croston method has two important disadvantages. First and foremost, not updating after (many) periods with zero demand renders the method unsuitable for dealing with obsolescence issues. Second, the method is positively biased and this is true for all points in time (i.e. considering the forecasts made at an arbitrary time period) and issue points only (i.e. considering the forecasts following a positive demand occurrence only). The second issue has been addressed in the literature by the proposal of an estimator (Syntetos–Boylan Approximation, SBA) that is approximately unbiased. In this paper, we propose a new method that overcomes both these shortcomings while not adding complexity. Different from the Croston method, the new method is unbiased (for all points in time) and it updates the demand probability instead of the demand interval, doing so in every period. The comparative merits of the new estimator are assessed by means of an extensive simulation experiment. The results indicate its superior performance and enable insights to be gained into the linkage between demand forecasting and obsolescence.

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## 1. Introduction

Intermittent demand appears sporadically, with some time periods showing no demand at all. When a demand occurs, the demand size may be constant or variable, perhaps highly so. Intermittent demand items may be any Stock Keeping Unit (SKU) within the range of products offered by an organization at any level of the supply chain. Such items may collectively account for up to 60% of the total stock value (Johnston et al., 2003) and are particularly prevalent in the aerospace, automotive, military and IT sectors. They are often the items at greatest risk of obsolescence (Porras and Dekker, 2008).

The occurrence of (many) periods with zero demand renders traditional forecasting techniques such as simple exponential smoothing or simple moving average unsuitable. Those methods ignore the fact that intermittent demand patterns are built from two elements: demand size and demand probability (or its inverse, the demand interval). The Croston (1972) method does differentiate between these two elements. Using exponential smoothing, the demand size and the demand interval are separately updated after each period with a positive demand. The ratio of the former over the latter then provides a forecast for the future demand per period.

The Croston method is often applied in practice (see, for example Fildes et al., 2008) and incorporated in Enterprise Resource Planning (ERP) type solutions such as SAP and specialised forecasting software packages such as Forecast Pro. However, the method has two important disadvantages. First, it is positively biased, as will be discussed further in the next section. Second and most important, by not updating after periods with zero demand, the forecasts become outdated after (many) periods with zero demand and unsuitable for estimating the risk of obsolescence. Obsolescence is an important topic for inventory management, especially for slow moving, intermittent demand. Molenaers (2010) discussed a case study where 54% of the parts stocked at a large petrochemical company had seen no demand for the last 5 years. Syntetos et al. (2009a) evaluated the inventory practices employed in the European spare parts logistics network of a Japanese manufacturer. One case was reported in Sweden where parts in stock had not 'moved' at all over the preceding 10 years. Hinton (1999) described that the US Department of Defence holds 60% excess of spare parts, with 18% of the parts (with a total value of $1.5 billion)

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having no demand at all. In such situations, decisions to remove 'dead stock' and reduce inventory levels are of great importance. Those decisions rely on accurate demand forecasts and associated risk of obsolescence estimates, especially after long intervals with zero demand.

Despite its importance, obsolescence is ill-researched. We refer interested readers to Teunter (1998, Section 2.2) for a review of obsolescence in the general area of inventory control, and to Kennedy et al. (2002, Section 3.5) for obsolescence in spare parts management. Recent contributions are those by Pinçe and Dekker (2011) and Van Jaarsveld and Dekker (2008). To the best of our knowledge, obsolescence has not been discussed in the forecasting literature. Fildes et al. (2008) and Syntetos et al. (2009b) who recently reviewed the link between forecasting and Operational Research (OR)/inventory planning, also did not discuss inventory obsolescence. Finally, recent attempts to link explicitly the intermittent nature of demand patterns to stock control (Syntetos et al., 2009c; Teunter et al., 2010) also did not take this issue into account.

In this paper, we propose a new forecasting method for intermittent demand that is always up-to-date and can deal with obsolescence as well as other inventory decisions. Different from the Croston method, which updates the demand interval, the new method updates the demand probability. This may appear as a small difference, since the demand interval is the inverse of the underlying probability of demand occurrence. However, the demand probability can be updated in every period, whereas the demand interval can only be updated after a positive demand. Moreover, as we will show in Section 3, using demand probability instead of demand interval eliminates the forecasting bias when considering an arbitrary point in time.

The new method achieves a high flexibility by using different smoothing constants for demand size and demand probability. We will derive an exact expression for the variance of the new method that is useful in setting the smoothing constants. Higher values make the new method more adaptable, but also increase its variance.

The performance of the new method (TSB for short – after Teunter, Syntetos and Babai) is evaluated through a simulation experiment on theoretically generated data. The experimental structure allows the evaluation of TSB's performance both on stationary and non-stationary demand data, the latter being generated in such a way so as to capture possible obsolescence-related scenarios. Performance is reported through the consideration of the Mean signed Error (ME – that expresses bias) and the Mean Squared Error (MSE – that is directly associated with the issue of variance).

Of course, the new method or in fact any other forecasting method can not prevent obsolescence completely. Indeed, obsolescence may be the outcome of bad management, where changes in the demand rate go unnoticed (for too long). However, with thousands of slow moving items in stock all too often is not easy at all to determine when an item should be discontinued. Having a forecasting method that automatically adjusts a forecast downward after (long) periods without demand, as the new method does, is certainly preferable with respect to (identifying) obsolescence. The same could be achieved by using a simple moving average (MA) or simple exponential smoothing (SES). Both methods are also unbiased when considering an arbitrary point in time. However, and as discussed above, the relevant forecasts are not built from constituent elements (sizes of demand and probability of demand occurrence). The explicit consideration of the probability of demand occurrence is a major determinant of the intuitive appeal and operational implications of the new estimator. With regards to the latter issue, a number of studies have established the fact that compound stock control models that rely upon both constituent elements perform considerably better than traditional stock control formulations that rely solely on estimates of mean demand (see, for example, Syntetos et al., 2009c; Teunter et al., 2010).

The remainder of this paper is organized as follows. In the next section, we review the most relevant literature. Section 3 introduces notations and formally defines the new forecasting method along with the demonstration of its unbiased nature and the derivation of its variance under stationary demand. A discussion on setting the smoothing constants for the new method is also included. In Section 4 the details governing our simulation experiment are outlined, after which the results are presented and discussed in Section 5. We end with conclusions and directions for further research in Section 6.

## 2. Research background

The standard method used in industry to forecast intermittent demand requirements is the Croston method (Croston, 1972). The method is incorporated in statistical forecasting software packages (e.g. Forecast Pro), and demand planning modules of component based enterprise and manufacturing solutions (e.g. Industrial and Financial Systems – IFS AB). It is also included in integrated real-time sales and operations planning processes (e.g. SAP Advanced Planning and Optimisation-APO 4.0).

Croston suggested separating intermittent demand series into their constituent elements, i.e. demand sizes and inter-demand intervals and separately updating their estimates through simple exponential smoothing (SES). He further suggested using the same value for the smoothing constant of both sizes and intervals, although the method may also be applied through the consideration of different smoothing constants (see also Schultz, 1987). The ratio of those estimates (sizes/intervals) may then be used as an estimate of the demand per period. The reciprocal of the inter-demand interval estimate is used essentially as the estimate of the probability of demand occurrence (that was modelled as following a Bernoulli sequence). However, Syntetos and Boylan (2001) showed Croston's method to suffer from a positive 'inversion bias' (for a random variable $X$: $1/E(X) \neq E(1/X)$) resulting in over-forecasting mean demand. Subsequently, they proposed a modification, the so-called Syntetos–Boylan Approximation or SBA for short (Syntetos and Boylan, 2005). The modified method applies a deflating factor $(1 - \alpha/2)$ that is linear in the smoothing constant (used for updating the inter-demand interval) to the Croston method. The method has been shown in a number of independent empirical studies to compare very favourably to Croston's estimator (see, for example, Eaves and Kingsman, 2004; Gutierrez et al., 2008). However, some (negative) bias remains and there are indeed cases when the SBA method may be more biased than the original Croston method (Wallström and Segerstedt, 2010; Teunter and Sani, 2009). Also, the deflating factor renders the method less intuitive, which may hinder its implementation in practice. The latter disadvantage applies even stronger to a more complicated modification proposed by Syntetos (2001) (see also Teunter and Sani, 2009), although that does remove the bias completely. Levén and Segerstedt (2004) also proposed a modified Croston method. Their method does not update the inter-demand interval and demand size separately, but does use the distinction under concern to directly update the demand per period. By doing so, this method avoids the original bias-causing issue. However, it is still biased and in fact even more heavily so than the original estimator (Boylan and Syntetos, 2007; Teunter and Sani, 2009).

Another disadvantage of the Croston method and all the above discussed variants is that the forecast is only updated after periods with positive demand. As a result, these methods are not up-to-date after (many) periods with zero demand and therefore cannot be used to estimate the risk of obsolescence and deal with the removal of excess/dead stock.

A method that is not specifically developed for intermittent demand, but does update the forecast after every period, is the naïve method that uses the last observed demand as the forecast for future periods. However, although this method is unbiased, it is too erratic for most practical situations. We remark that our new estimator contains the naïve method as a special case (when both smoothing constants are set to one), but that its variance can be controlled by using different values for the smoothing constants.

For completeness, we should also refer to some other studies conducted in the area of intermittent demand forecasting. Altay et al. (2008) considered a modification of Holt's double exponential smoothing for forecasting intermittent demands which was found to perform well under a customer service level constraint. Also, it is well known that service parts are typically characterized by intermittent demand patterns and hence the current discussion is relevant for maintenance management. In such a context, causal methods have also been shown to have an important role (Ghobbar and Friend, 2002, 2003). Syntetos et al. (2009d) considered the effect of integrating judgemental adjustments into intermittent demand forecasts produced by a software solution. Finally, a number of non-parametric and semi-parametric approaches have also been discussed in the literature (e.g. Porras and Dekker, 2008; Snyder, 2002; Teunter and Duncan, 2009; Willemain et al., 2004). Such methods typically reply upon bootstrapping procedures and although recent advancements in computing render their application in practice a straightforward exercise, more empirical studies are required in order to develop our understanding on how these methods compare to parametric Croston-type estimators. At this point, we should mention that none of the above studies have addressed the issue of obsolescence.

## 3. Notation, mathematical description of the new method and discussion

We introduce the following notation:

$Y_t$: Demand for an item in period $t$.
$Y'_t$: Estimate of mean demand per period at the end of period t for period $t+1$.
$z_t$: Actual demand size in period $t$.
$z'_t$: Estimate of mean demand size at the end of period t.
$p_t$: Demand occurrence indicator for period t, so that: $p_t = \begin{cases} 1 & \text{if demand occurrs at time t (i.e. } Y_t > 0) \\ 0 & \text{otherwise} \end{cases}$.
$p'_t$: Estimate of the probability of a demand occurrence at the end of period $t$.
$\alpha, \beta$: Smoothing constants ($0 \leqslant \alpha, \beta \leqslant 1$).

### 3.1. New method

The new method uses separate simple exponentially smoothed estimates of the demand probability and the demand size. The estimate of the probability of occurrence is updated every time period. The estimate of the demand size is only updated at the end of periods with positive demand. The product of the estimates for demand size and demand probability provides the forecast of the demand per period. We use two different smoothing constants, particularly because the demand probability is updated more often than the demand size

$$\text{If} \quad p_t = 0 : p'_t = p'_{t-1} + \beta(0 - p'_{t-1}), \quad z'_t = z'_{t-1}, \quad Y'_t = p'_t z'_t,$$

$$\text{If} \quad p_t = 1 : p'_t = p'_{t-1} + \beta(1 - p'_{t-1}), \quad z'_t = z'_{t-1} + \alpha(z_t - z'_{t-1}), \quad Y'_t = p'_t z'_t$$

It is easy to see that for the special case with $\alpha = \beta = 1$, the new method gives $Y'_t = 0$ if $p_t = 0$ and $Y'_t = Y_t$ if $p_t = 1$. So the new method is identical to the naïve method for $\alpha = \beta = 1$.

Note that in estimating the demand probability, the TSB method assigns the same weight to the most recent, say the last 24, observations, regardless of the number of periods in which demand was positive. For instance, when $\beta = 0.2$, approximately 99.6% weight ($\sum_{i=0}^{24} \beta(1-\beta)^i$) is assigned to these observations. In contrast, for the Croston and SBA methods using a smoothing constant of $\alpha = 0.2$ amounts to assigning 99.6% weight to the 25 most recent periods with positive demand. So Croston's method automatically uses a longer demand history (in time periods) for parts that are slow moving if $\beta = \alpha$. Therefore, it may be better to set $\beta < \alpha$. Moreover, the 'optimal' value for $\beta$ (relative to $\alpha$) may depend on the demand rate. Theoretical results in Section 3.3 and numerical results in Section 5 will confirm these expectations and provide some guidelines on how to set $\alpha$ and $\beta$.

### 3.2. Bias and variance under stationary demand

Bias and variance are important performance measures for forecasting methods. In this sub-section, we will prove that the new method is unbiased and derive an exact expression for its variance under the assumption of stationary demand (demand probability $p$: $E(p_t) = p$, $Var(p_t) = p(1-p)$, mean demand size $\mu$, standard deviation of demand size $\sigma$) with independence of demand occurrences in different periods. We remark that stationary demand is not all that matters, and will discuss this further in the next sub-section.

According to standard statistical results:

$$E[Y'_t] = E[p'_t]\, E[z'_t] \tag{1}$$

and

$$E\left[(Y'_t)^2\right] = E\left[(p'_t)^2\right] E\left[(z'_t)^2\right] = (V[p'_t] + (E[p'_t])^2)\,(V[z'_t] + (E[z'_t])^2). \tag{2}$$

Since $p'_t$ can be written (see e.g. Axsäter, 2006) as a weighted average of Bernoulli distributed variables $p_t, p_{t-1}, p_{t-2}, \ldots$ with weights: $\beta, \beta(1-\beta), \beta(1-\beta)^2, \ldots$ it holds that (see Appendix for detailed derivations)

$$E[p'_t] = \sum_{i=0}^{\infty} \beta(1-\beta)^i p = p \tag{3}$$

and

$$V[p'_t] = \sum_{i=0}^{\infty} (\beta(1-\beta)^i)^2 p(1-p) = p(1-p)\frac{\beta}{2-\beta}. \tag{4}$$

Similarly, we get

$$E[z'_t] = \mu \tag{5}$$

and

$$Var[z'_t] = \sigma^2 \frac{\alpha}{2-\alpha} \tag{6}$$

Combining (1)–(6) gives

$$E[Y'_t] = p\mu$$

and

$$Var[Y'_t] = E[(Y'_t)^2] - (E[Y'_t])^2 = \left(p(1-p)\frac{\beta}{2-\beta} + p^2\right)\left(\sigma^2\frac{\alpha}{2-\alpha} + \mu^2\right) - (p\mu)^2 = \frac{\alpha}{2-\alpha}\frac{\beta}{2-\beta}\sigma^2 p(1-p) + \frac{\alpha}{2-\alpha}\sigma^2 p^2 + \frac{\beta}{2-\beta}\mu^2 p(1-p). \tag{7}$$

Please note that although the TSB method is unbiased when all points in time are considered, the method suffers from a statistical bias for issue points only (i.e. the estimates isolated in demand occurring periods only). Conceptually, the bias generation process is similar to that characterising SES when applied directly to intermittent demand data.

### 3.3. Smoothing constants

A common objective in forecasting/statistics is to find the minimum variance, unbiased estimator. However, this can only be achieved if the underlying demand process is known. In practice, especially for intermittent demand, this is typically not the case. Indeed, it is often unclear whether the demand process is stationary or non-stationary. If demand is suspected to be highly non-stationary, then a forecasting method that adapts quickly is preferable. For our new method, this is achieved by choosing 'sufficiently large' smoothing constants. If this is not the case, then small smoothing constants are preferable.

The variance expression (7) for the new method under the assumption of stationary demand, which was derived in the previous sub-section, provides some guidance in setting the smoothing constants. If demand may be (close to) stationary, then the outcome of (7) should not be 'too large'. It is obvious from (7) that this is achieved by choosing small enough smoothing constants. Indeed (7) shows that a zero variance under stationary demand can be achieved (in the long run) by setting $\alpha = \beta = 0$. On the other hand, the forecast should be sufficiently adaptive in case demand turns out to be highly non-stationary, and so the smoothing constants should not be too small either.

One way of setting the smoothing constants without knowing the underlying demand process is through empirical optimisation based on the demand history, although this is complicated for intermittent demand because of the typical lack of sufficient data. Another option is to use fixed values. For simple exponential smoothing as well as for the Croston method, values between 0.05 and 0.2 are usually recommended (Syntetos and Boylan, 2005). However, it is important to keep in mind that for the TSB method, the values of the smoothing constants should depend on the period length. For spare parts, for instance, values in the above mentioned range could be suitable for monthly periods, but much smaller values are needed for periods of say one day.

A comparison of the TSB variance in (7) to that of the Croston method (Syntetos et al., 2005) shows that the former is at most the latter if $\beta/(2-\beta) \leqslant p\alpha/(2-\alpha)$, or equivalently, $\beta \leqslant 2p\alpha/[2-\alpha(1-p)]$. Since smoothing constants are usually set to small values, this condition is almost equivalent to $\beta \leqslant p\alpha$.. This simplified condition can be used as a guideline for achieving a variance reduction for TSB as compared to Croston, which also implies a reduction of the Mean Square Error.

### 3.4. Linkage to obsolescence

Different from the Croston and SBA methods that update the estimates only after a demand occurrence, the new method updates the demand probability in every time period. The probability is adjusted upward in periods with positive demand and downward in periods with zero demand. After many periods with zero demand and therefore many downwards adjustments, the TSB estimate for the probability of having positive demand approaches zero, and therefore the estimate of the mean demand per period approaches zero as well.

In this way, the TSB method is able to quickly react to situations with sudden obsolescence or with an increased risk of obsolescence. If an item suddenly becomes obsolete (e.g. for a spare part because equipment is taken out of operation) and demands no longer occur as a result, then the TSB method will adjust its estimates to zero as described above. The Croston and SBA method, on the contrary, are not able to react and will stick to the pre-obsolescence estimates. In 'less extreme' situations where the demand rate decreases (e.g. because a competitor enters the market), the Croston and SBA methods do react. However, the adjustment does not happen until positive demands occur, and hence will be slow especially if demand intervals are long. The TSB method responds more quickly by adjusting estimates in every period.

Therefore, we expect the TSB method to provide more accurate forecasts in cases with sudden obsolescence or with an increased risk of obsolescence. This will be confirmed by our numerical results in Section 5. In our numerical investigation, we also include Simple Exponential Smoothing (SES). Although this forecasting method has important shortcomings for intermittent demand (as separate estimates for demand probability and size of demand are not obtained, although those are essential for inventory control), it does respond quickly to situations with sudden obsolescence or decreasing demand.

## 4. Numerical investigation

In this section we provide details on the experimental structure employed for the purposes of our investigation, followed by the results and a discussion in Section 5. The simulation programme is written in Visual Basic (embedded in Excel) and the details discussed below should facilitate a straightforward replication of our results should one wish to do so.

For the purposes of our investigation, demand sizes were assumed to be stationary following the logarithmic distribution. Other distributions could have been considered (e.g. Lognormal – Willemain et al., 1994; Gamma – Teunter et al., 2010; truncated Poisson or truncated Pascal – Syntetos, 2001, etc). We chose the logarithmic for the following reasons. First, it is discrete (as real demand is too). Second, its flexibility, as expressed through the variance to mean ratio allowed us to model from very low demand patterns (demand size equal to 1) to very lumpy demands (considerably variable demand sizes). Third, the logarithmic distribution has some theoretical justification in its support. Poisson demand arrivals coupled with logarithmically distributed demand sizes result in negative binomial distributed (NBD) demand per time period, which has received a lot of attention in the context of representing intermittent demands and has considerable empirical evidence in its support (e.g. Kwan, 1991). Although, demand arrivals are modelled as Bernoulli for the purposes of our work, this can be interpreted as the discrete analogue of a Poisson process.

The Logarithmic distribution is characterized by the shape parameter $c$, $0 < c < 1$, and if we let $k = -1/\log(1-c)$ then the probability distribution is defined as $kc^x/x$ for $x \geqslant 1$ (and integer). Logarithmic sizes were generated under two experimental conditions: logarithmic parameter $c = 0.001$ ($\mu = 1$, $\sigma = 0$) and $c = 0.9$ ($\mu = 4$, $\sigma = 5$). In the former case, very slow moving demand (with unit-sized transactions) is simulated whereas in the latter case, we generate very lumpy demand patterns. The theoretical demand size distribution under $c = 0.9$ is indicated in Fig. 1.

Demand data is generated for 10 years (120 months/periods); sampling variation is dealt with through 10 replications of each control parameter combination. With regards to the Bernoulli demand occurrence distribution, three cases are considered (the last two reflect the issue of obsolescence).

- Stationary demand: constant demand probability $p = 0.2$ or $p = 0.5$.
- Linearly decreasing demand: demand probability starts at $p_0 = 0.2$ or $p_0 = 0.5$ in the first period, and decreases linearly to 0 in the last period.

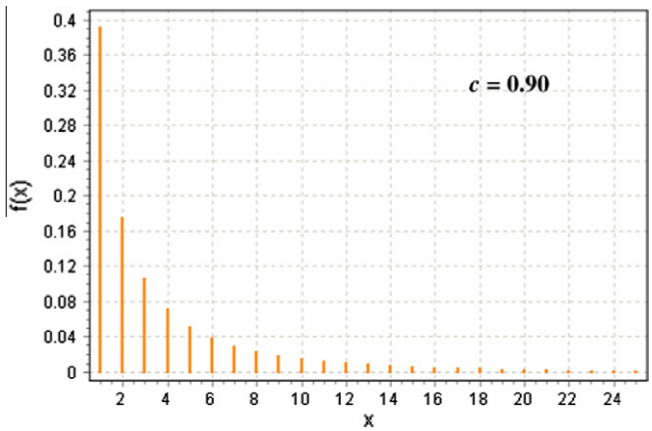


**Fig. 1.** Logarithmic distribution with parameter $c = 0.90$ and density function $f(\cdot)$.

- Sudden Obsolescence: after five years of stationary demand with demand probability $p = 0.2$ or $p = 0.5$, the demand probability suddenly drops to 0.

Four estimators are considered for the purpose of our numerical investigation: SBA, Croston, SES, and TSB. The smoothing constant employed by SES, Croston and SBA (for updating both sizes and intervals) and TSB (for updating sizes only) is set to $\alpha = 0.1$, 0.2 or 0.3. The smoothing constant used for updating the demand probability in the TSB estimator is set to $\beta = 0.01$, 0.02, 0.03, 0.04, 0.05, 0.1, 0.2, or 0.3. Small values for $\beta$ (i.e. $\beta = 0.01$, 0.02, 0.03, 0.04, 0.05) are included since, as discussed in the previous section, it is suspected that such values provide a better performance for the TSB estimator (for stationary demand). Performance for all methods is evaluated from the very first period and (for fairness) all methods start with the correct initial values.

With regards to performance measurement, two forecast accuracy measures are used: the Mean signed Error (ME) as a measure of bias, and the Mean Squared Error (MSE) that relates to the issue of variance (of the estimates). As discussed above, ten replications are conducted for each experimental scenario (combination of control parameters); the ME and MSE results reported in the next section constitute averages across those replications.

The Mean (Signed) Error (ME) for any of the estimators considered in our numerical study (and for a particular control parameter combination) is calculated as follows:

$$\mathrm{ME} = \frac{\sum_{i=1}^{k}\sum_{t=0}^{n-1}(Y'_t - Y_{t+1})}{nk}, \tag{8}$$

where: $n$ is the simulation length (10,000), $Y'_0$ is the population expected value for the control parameter combination under concern and $k$ is the number of replications (equal to 10).

Similarly for the MSE:

$$\mathrm{MSE} = \frac{\sum_{i=1}^{k}\sum_{t=0}^{n-1}(Y'_t - Y_{t+1})^2}{nk}. \tag{9}$$

## 5. Numerical results

The results for stationary demand, linearly decreasing demand and sudden obsolescence will first be discussed separately in Sections 5.1–5.3, respectively. Each of these sub-sections starts with tabulated results (organised in four tables, one for each of

**Table 1**
ME and MSE results – stationary demand ($c = 0.001$; $p = 0.2$).

| Estimator | ME | | | | MSE | | | |
|---|---|---|---|---|---|---|---|---|
| | α Smoothing constant | | | β | α Smoothing constant | | | β |
| | 0.10 | 0.20 | 0.30 | | 0.10 | 0.20 | 0.30 | |
| SBA | −0.0089 | −0.0088 | −0.0120 | | 0.1763 | 0.1774 | 0.1788 | |
| Croston | 0.0026 | 0.0155 | 0.0261 | | 0.1763 | 0.1782 | 0.1810 | |
| SES | −0.0022 | −0.0013 | −0.0006 | | 0.1840 | 0.1945 | 0.2061 | |
| TSB | −0.0150 | −0.0150 | −0.0150 | 0.01 | 0.1760 | 0.1760 | 0.1760 | 0.01 |
| | −0.0090 | −0.0090 | −0.0090 | 0.02 | 0.1766 | 0.1766 | 0.1766 | 0.02 |
| | −0.0061 | −0.0061 | −0.0061 | 0.03 | 0.1774 | 0.1774 | 0.1774 | 0.03 |
| | −0.0045 | −0.0045 | −0.0045 | 0.04 | 0.1783 | 0.1783 | 0.1783 | 0.04 |
| | −0.0036 | −0.0036 | −0.0036 | 0.05 | 0.1792 | 0.1792 | 0.1792 | 0.05 |
| | −0.0022 | −0.0022 | −0.0022 | 0.10 | 0.1840 | 0.1840 | 0.1840 | 0.10 |
| | −0.0013 | −0.0013 | −0.0013 | 0.20 | 0.1945 | 0.1945 | 0.1945 | 0.20 |
| | −0.0006 | −0.0006 | −0.0006 | 0.30 | 0.2061 | 0.2061 | 0.2061 | 0.30 |

**Table 2**
ME and MSE results – stationary demand ($c = 0.90$; $p = 0.2$).

| Estimator | ME | | | | MSE | | | |
|---|---|---|---|---|---|---|---|---|
| | α Smoothing constant | | | β | α Smoothing constant | | | β |
| | 0.10 | 0.20 | 0.30 | | 0.10 | 0.20 | 0.30 | |
| SBA | −0.0167 | −0.0418 | −0.0651 | | 8.3531 | 8.4167 | 8.4958 | |
| Croston | 0.0252 | 0.0440 | 0.0669 | | 8.3609 | 8.4561 | 8.5989 | |
| SES | −0.0046 | 0.0017 | 0.0038 | | 8.6066 | 9.0464 | 9.5465 | |
| TSB | −0.0323 | −0.0619 | −0.0811 | 0.01 | 8.3304 | 8.3436 | 8.3725 | 0.01 |
| | −0.0190 | −0.0477 | −0.0670 | 0.02 | 8.3495 | 8.3660 | 8.3962 | 0.02 |
| | −0.0108 | −0.0383 | −0.0573 | 0.03 | 8.3679 | 8.3886 | 8.4214 | 0.03 |
| | −0.0052 | −0.0314 | −0.0499 | 0.04 | 8.3842 | 8.4095 | 8.4458 | 0.04 |
| | −0.0010 | −0.0259 | −0.0436 | 0.05 | 8.3986 | 8.4288 | 8.4691 | 0.05 |
| | 0.0111 | −0.0078 | −0.0213 | 0.10 | 8.4564 | 8.5099 | 8.5735 | 0.10 |
| | 0.0199 | 0.0069 | −0.0008 | 0.20 | 8.5699 | 8.6564 | 8.7634 | 0.20 |
| | 0.0224 | 0.0116 | 0.0064 | 0.30 | 8.7022 | 8.8072 | 8.9474 | 0.30 |

**Table 3**
ME and MSE results – stationary demand ($c = 0.001$; $p = 0.5$).

| Estimator | ME | | | | MSE | | | |
|---|---|---|---|---|---|---|---|---|
| | α Smoothing constant | | | β | α Smoothing constant | | | β |
| | 0.10 | 0.20 | 0.30 | | 0.10 | 0.20 | 0.30 | |
| SBA | −0.0130 | −0.0263 | −0.0406 | | 0.2562 | 0.2615 | 0.2662 | |
| Croston | 0.0129 | 0.0269 | 0.0413 | | 0.2567 | 0.2639 | 0.2720 | |
| SES | −0.0001 | 0.0009 | 0.0014 | | 0.2627 | 0.2760 | 0.2911 | |
| TSB | −0.0022 | −0.0022 | −0.0022 | 0.01 | 0.2521 | 0.2521 | 0.2522 | 0.01 |
| | −0.0014 | −0.0014 | −0.0013 | 0.02 | 0.2530 | 0.2531 | 0.2531 | 0.02 |
| | −0.0011 | −0.0010 | −0.0010 | 0.03 | 0.2541 | 0.2541 | 0.2542 | 0.03 |
| | −0.0009 | −0.0008 | −0.0008 | 0.04 | 0.2552 | 0.2553 | 0.2553 | 0.04 |
| | −0.0008 | −0.0007 | −0.0007 | 0.05 | 0.2564 | 0.2565 | 0.2565 | 0.05 |
| | −0.0002 | −0.0001 | −0.0001 | 0.10 | 0.2627 | 0.2627 | 0.2628 | 0.10 |
| | 0.0009 | 0.0010 | 0.0010 | 0.20 | 0.2758 | 0.2759 | 0.2760 | 0.20 |
| | 0.0014 | 0.0015 | 0.0015 | 0.30 | 0.2907 | 0.2908 | 0.2910 | 0.30 |

the combinations of the 2 coefficients of the logarithmic distribution and the 2 underlying probabilities of demand occurrence considered), followed by a discussion. Section 5.4 wraps up with a summary of all the results.

### 5.1. Stationary demand

Note from Tables 1 and 3 that for cases with very slow moving demand and almost always unit-sized transactions ($c = 0.001$), the performance of TSB does not depend on α. This is explained by the fact that TSB only uses α to update the demand size estimate, but that estimate stays fixed at 1 whatever the value of α if all demands are of size 1. For the other methods, α does affect the estimate for the demand per period, directly for SES and via the demand interval estimate for Croston and SBA.

Overall, we see from Tables 1–4 that TSB and SES are less biased than Croston and SBA. This is expected, as the first two methods are theoretically known to be unbiased when all points in time are considered, whereas the latter two are not. Indeed, the exceptions where TSB and/or SES do not have a lower bias may be explained in terms of numerical inaccuracy.

The MSE performance of SES is very poor. This also confirms our expectations, as this method ignores the intermittent character of the demand. TSB provides the best MSE performance, but only if β

**Table 4**
ME and MSE results – stationary demand ($c = 0.90$; $p = 0.5$).

| Estimator | ME | | | | MSE | | | |
|---|---|---|---|---|---|---|---|---|
| | α Smoothing constant | | | β | α Smoothing constant | | | β |
| | 0.10 | 0.20 | 0.30 | | 0.10 | 0.20 | 0.30 | |
| SBA | −0.0189 | −0.0844 | −0.1558 | | 15.5384 | 15.9597 | 16.3440 | |
| Croston | 0.0797 | 0.1165 | 0.1506 | | 15.5833 | 16.1457 | 16.7736 | |
| SES | 0.0197 | 0.0128 | 0.0101 | | 15.9530 | 16.8331 | 17.7972 | |
| TSB | 0.0180 | 0.0065 | −0.0073 | 0.01 | 15.4849 | 15.7894 | 16.0838 | 0.01 |
| | 0.0212 | 0.0103 | −0.0032 | 0.02 | 15.5045 | 15.8180 | 16.1209 | 0.02 |
| | 0.0242 | 0.0140 | 0.0010 | 0.03 | 15.5226 | 15.8477 | 16.1613 | 0.03 |
| | 0.0258 | 0.0164 | 0.0039 | 0.04 | 15.5409 | 15.8780 | 16.2026 | 0.04 |
| | 0.0263 | 0.0176 | 0.0056 | 0.05 | 15.5601 | 15.9083 | 16.2434 | 0.05 |
| | 0.0219 | 0.0156 | 0.0058 | 0.10 | 15.6687 | 16.0532 | 16.4285 | 0.10 |
| | 0.0144 | 0.0096 | 0.0024 | 0.20 | 15.9200 | 16.3258 | 16.7393 | 0.20 |
| | 0.0111 | 0.0067 | 0.0009 | 0.30 | 16.2036 | 16.6191 | 17.0552 | 0.30 |

**Table 5**
ME and MSE results – decreasing demand ($c = 0.001$; $p = 0.2$).

| Estimator | ME | | | | MSE | | | |
|---|---|---|---|---|---|---|---|---|
| | α Smoothing constant | | | β | α Smoothing constant | | | β |
| | 0.10 | 0.20 | 0.30 | | 0.10 | 0.20 | 0.30 | |
| SBA | 0.0672 | 0.0535 | 0.0452 | | 0.0977 | 0.0966 | 0.0968 | |
| Croston | 0.0763 | 0.0712 | 0.0718 | | 0.0990 | 0.0990 | 0.1008 | |
| SES | 0.0158 | 0.0083 | 0.0056 | | 0.0959 | 0.1009 | 0.1063 | |
| TSB | 0.0669 | 0.0669 | 0.0669 | 0.01 | 0.0970 | 0.0970 | 0.0970 | 0.01 |
| | 0.0506 | 0.0506 | 0.0506 | 0.02 | 0.0948 | 0.0948 | 0.0948 | 0.02 |
| | 0.0402 | 0.0402 | 0.0402 | 0.03 | 0.0940 | 0.0940 | 0.0940 | 0.03 |
| | 0.0331 | 0.0331 | 0.0331 | 0.04 | 0.0938 | 0.0938 | 0.0938 | 0.04 |
| | 0.0281 | 0.0281 | 0.0281 | 0.05 | 0.0939 | 0.0939 | 0.0939 | 0.05 |
| | 0.0158 | 0.0158 | 0.0158 | 0.10 | 0.0959 | 0.0959 | 0.0959 | 0.10 |
| | 0.0083 | 0.0083 | 0.0083 | 0.20 | 0.1009 | 0.1009 | 0.1009 | 0.20 |
| | 0.0056 | 0.0056 | 0.0056 | 0.30 | 0.1063 | 0.1063 | 0.1063 | 0.30 |

**Table 6**
ME and MSE results – decreasing demand ($c = 0.90$; $p = 0.2$).

| Estimator | ME | | | | MSE | | | |
|---|---|---|---|---|---|---|---|---|
| | $\alpha$ Smoothing constant | | | $\beta$ | $\alpha$ Smoothing constant | | | $\beta$ |
| | 0.10 | 0.20 | 0.30 | | 0.10 | 0.20 | 0.30 | |
| SBA | 0.3039 | 0.2120 | 0.1607 | | 2.6889 | 2.6634 | 2.6709 | |
| Croston | 0.3354 | 0.2683 | 0.2411 | | 2.7106 | 2.7004 | 2.7294 | |
| SES | 0.0639 | 0.0327 | 0.0221 | | 2.7120 | 2.8578 | 3.0211 | |
| TSB | 0.3157 | 0.2840 | 0.2636 | 0.01 | 2.6925 | 2.6992 | 2.7172 | 0.01 |
| | 0.2427 | 0.2159 | 0.1985 | 0.02 | 2.6508 | 2.6580 | 2.6755 | 0.02 |
| | 0.1979 | 0.1739 | 0.1582 | 0.03 | 2.6366 | 2.6432 | 2.6597 | 0.03 |
| | 0.1687 | 0.1465 | 0.1319 | 0.04 | 2.6332 | 2.6393 | 2.6548 | 0.04 |
| | 0.1486 | 0.1277 | 0.1138 | 0.05 | 2.6345 | 2.6402 | 2.6552 | 0.05 |
| | 0.1025 | 0.0852 | 0.0736 | 0.10 | 2.6588 | 2.6648 | 2.6804 | 0.10 |
| | 0.0770 | 0.0623 | 0.0526 | 0.20 | 2.7177 | 2.7279 | 2.7499 | 0.20 |
| | 0.0682 | 0.0542 | 0.0451 | 0.30 | 2.7850 | 2.7987 | 2.8265 | 0.30 |

is sufficiently small. This numerically confirms the guideline for setting $\beta$ that was discussed in Section 3.3 (see Tables 5–12).

### 5.2. Linearly decreasing demand (increasing risk of obsolescence)

None of the methods directly incorporates the downward trend in demand, and as a result all are positively biased. As was observed for stationary demand, Croston and SBA are the most biased. These methods update only when there is a positive demand, leading to a slow adjustment to the decrease in demand. TSB and SES do update in every period, leading to smaller biases provided that the smoothing constants are sufficiently large.

Croston, SBA and TSB have similar MSEs, and those are almost unaffected by the smoothing constants. The TSB method does not appear to have much of on advantage over SBA with respect to MSE, except for a few values of $\beta$. SES is somewhat less accurate, especially when the smoothing constant is large as this leads to erratic forecasts.

### 5.3. Step-changed demand (sudden obsolescence)

Again, TSB and SES are less biased than Croston and SBA. Indeed, the relative performance differences are much larger than those reported for the previous cases with stationary and linearly decreasing demand. This is explained by the inability of Croston and SBA to react to sudden obsolescence, whereas the occurrence of positive demands did allow adjustment, albeit at a slow pace, for the previous cases. So, the TSB method will obviously outperform Croston or

**Table 7**
ME and MSE results – decreasing demand ($c = 0.001$; $p = 0.5$)

| Estimator | ME | | | | MSE | | | |
|---|---|---|---|---|---|---|---|---|
| | $\alpha$ Smoothing constant | | | $\beta$ | $\alpha$ Smoothing constant | | | $\beta$ |
| | 0.10 | 0.20 | 0.30 | | 0.10 | 0.20 | 0.30 | |
| SBA | 0.0873 | 0.0517 | 0.0311 | | 0.1992 | 0.1956 | 0.1962 | |
| Croston | 0.1072 | 0.0900 | 0.0882 | | 0.2028 | 0.2013 | 0.2047 | |
| SES | 0.0374 | 0.0197 | 0.0134 | | 0.1971 | 0.2073 | 0.2196 | |
| TSB | 0.1516 | 0.1516 | 0.1516 | 0.01 | 0.2179 | 0.2179 | 0.2179 | 0.01 |
| | 0.1166 | 0.1166 | 0.1166 | 0.02 | 0.2048 | 0.2048 | 0.2048 | 0.02 |
| | 0.0935 | 0.0935 | 0.0935 | 0.03 | 0.1990 | 0.1990 | 0.1990 | 0.03 |
| | 0.0775 | 0.0775 | 0.0775 | 0.04 | 0.1963 | 0.1963 | 0.1963 | 0.04 |
| | 0.0660 | 0.0660 | 0.0660 | 0.05 | 0.1952 | 0.1952 | 0.1952 | 0.05 |
| | 0.0373 | 0.0373 | 0.0373 | 0.10 | 0.1971 | 0.1971 | 0.1971 | 0.10 |
| | 0.0197 | 0.0197 | 0.0197 | 0.20 | 0.2073 | 0.2073 | 0.2073 | 0.20 |
| | 0.0134 | 0.0134 | 0.0134 | 0.30 | 0.2196 | 0.2196 | 0.2196 | 0.30 |

**Table 8**
ME and MSE results – decreasing demand ($c = 0.90$; $p = 0.5$).

| Estimator | ME | | | | MSE | | | |
|---|---|---|---|---|---|---|---|---|
| | $\alpha$ Smoothing constant | | | $\beta$ | $\alpha$ Smoothing constant | | | $\beta$ |
| | 0.10 | 0.20 | 0.30 | | 0.10 | 0.20 | 0.30 | |
| SBA | 0.4250 | 0.2680 | 0.1841 | | 8.5839 | 8.6347 | 8.7239 | |
| Croston | 0.4893 | 0.3863 | 0.3573 | | 8.6594 | 8.7895 | 9.0010 | |
| SES | 0.1612 | 0.0828 | 0.0556 | | 8.6543 | 9.0769 | 9.5600 | |
| TSB | 0.7088 | 0.6859 | 0.6730 | 0.01 | 8.9642 | 9.2291 | 9.5383 | 0.01 |
| | 0.5290 | 0.5083 | 0.4966 | 0.02 | 8.6729 | 8.8792 | 9.1136 | 0.02 |
| | 0.4185 | 0.3993 | 0.3883 | 0.03 | 8.5481 | 8.7256 | 8.9208 | 0.03 |
| | 0.3459 | 0.3280 | 0.3174 | 0.04 | 8.4873 | 8.6503 | 8.8242 | 0.04 |
| | 0.2954 | 0.2786 | 0.2685 | 0.05 | 8.4548 | 8.6107 | 8.7731 | 0.05 |
| | 0.1775 | 0.1646 | 0.1574 | 0.10 | 8.4147 | 8.5709 | 8.7282 | 0.10 |
| | 0.1094 | 0.0998 | 0.0963 | 0.20 | 8.4528 | 8.6271 | 8.8094 | 0.20 |
| | 0.0848 | 0.0758 | 0.0738 | 0.30 | 8.5423 | 8.7271 | 8.9269 | 0.30 |

**Table 9**
ME and MSE results – sudden obsolescence ($c = 0.001$; $p = 0.2$).

| Estimator | ME | | | | MSE | | | |
|---|---|---|---|---|---|---|---|---|
| | α Smoothing constant | | | β | α Smoothing constant | | | β |
| | 0.10 | 0.20 | 0.30 | | 0.10 | 0.20 | 0.30 | |
| SBA | 0.1040 | 0.1034 | 0.0995 | | 0.1057 | 0.1068 | 0.1070 | |
| Croston | 0.1149 | 0.1264 | 0.1355 | | 0.1082 | 0.1127 | 0.1169 | |
| SES | 0.0168 | 0.0084 | 0.0056 | | 0.0888 | 0.0929 | 0.0985 | |
| TSB | 0.0746 | 0.0746 | 0.0746 | 0.01 | 0.0951 | 0.0951 | 0.0951 | 0.01 |
| | 0.0586 | 0.0586 | 0.0586 | 0.02 | 0.0914 | 0.0914 | 0.0914 | 0.02 |
| | 0.0470 | 0.0470 | 0.0470 | 0.03 | 0.0896 | 0.0896 | 0.0896 | 0.03 |
| | 0.0384 | 0.0384 | 0.0384 | 0.04 | 0.0887 | 0.0887 | 0.0887 | 0.04 |
| | 0.0321 | 0.0321 | 0.0321 | 0.05 | 0.0882 | 0.0882 | 0.0882 | 0.05 |
| | 0.0168 | 0.0168 | 0.0168 | 0.10 | 0.0888 | 0.0888 | 0.0888 | 0.10 |
| | 0.0084 | 0.0084 | 0.0084 | 0.20 | 0.0929 | 0.0929 | 0.0929 | 0.20 |
| | 0.0056 | 0.0056 | 0.0056 | 0.30 | 0.0985 | 0.0985 | 0.0985 | 0.30 |

**Table 10**
ME and MSE results – sudden obsolescence ($c = 0.90$; $p = 0.2$).

| Estimator | ME | | | | MSE | | | |
|---|---|---|---|---|---|---|---|---|
| | α Smoothing constant | | | β | α Smoothing constant | | | β |
| | 0.10 | 0.20 | 0.30 | | 0.10 | 0.20 | 0.30 | |
| SBA | 0.3857 | 0.3508 | 0.3114 | | 4.5714 | 4.5988 | 4.6357 | |
| Croston | 0.4286 | 0.4376 | 0.4423 | | 4.6127 | 4.7003 | 4.8202 | |
| SES | 0.0671 | 0.0336 | 0.0224 | | 4.4121 | 4.6454 | 4.9053 | |
| TSB | 0.2861 | 0.2576 | 0.2312 | 0.01 | 4.4133 | 4.4032 | 4.4016 | 0.01 |
| | 0.2340 | 0.2118 | 0.1903 | 0.02 | 4.3623 | 4.3643 | 4.3719 | 0.02 |
| | 0.1945 | 0.1770 | 0.1591 | 0.03 | 4.3355 | 4.3450 | 4.3586 | 0.03 |
| | 0.1651 | 0.1509 | 0.1357 | 0.04 | 4.3216 | 4.3364 | 4.3544 | 0.04 |
| | 0.1430 | 0.1314 | 0.1181 | 0.05 | 4.3148 | 4.3337 | 4.3554 | 0.05 |
| | 0.0882 | 0.0833 | 0.0753 | 0.10 | 4.3187 | 4.3512 | 4.3864 | 0.10 |
| | 0.0576 | 0.0564 | 0.0517 | 0.20 | 4.3747 | 4.4230 | 4.4767 | 0.20 |
| | 0.0469 | 0.0467 | 0.0431 | 0.30 | 4.4539 | 4.5139 | 4.5823 | 0.30 |

**Table 11**
ME and MSE results – sudden obsolescence ($c = 0.001$; $p = 0.5$).

| Estimator | ME | | | | MSE | | | |
|---|---|---|---|---|---|---|---|---|
| | α Smoothing constant | | | β | α Smoothing constant | | | β |
| | 0.10 | 0.20 | 0.30 | | 0.10 | 0.20 | 0.30 | |
| SBA | 0.2360 | 0.2174 | 0.2000 | | 0.2460 | 0.2390 | 0.2339 | |
| Croston | 0.2617 | 0.2695 | 0.2798 | | 0.2592 | 0.2658 | 0.2757 | |
| SES | 0.0420 | 0.0210 | 0.0140 | | 0.1421 | 0.1440 | 0.1502 | |
| TSB | 0.1888 | 0.1888 | 0.1888 | 0.01 | 0.1995 | 0.1995 | 0.1995 | 0.01 |
| | 0.1471 | 0.1471 | 0.1471 | 0.02 | 0.1746 | 0.1746 | 0.1746 | 0.02 |
| | 0.1174 | 0.1174 | 0.1174 | 0.03 | 0.1614 | 0.1614 | 0.1614 | 0.03 |
| | 0.0959 | 0.0959 | 0.0959 | 0.04 | 0.1539 | 0.1539 | 0.1539 | 0.04 |
| | 0.0801 | 0.0801 | 0.0801 | 0.05 | 0.1493 | 0.1493 | 0.1493 | 0.05 |
| | 0.0419 | 0.0419 | 0.0419 | 0.10 | 0.1421 | 0.1421 | 0.1421 | 0.10 |
| | 0.0210 | 0.0210 | 0.0210 | 0.20 | 0.1440 | 0.1440 | 0.1440 | 0.20 |
| | 0.0140 | 0.0140 | 0.0140 | 0.30 | 0.1502 | 0.1502 | 0.1502 | 0.30 |

SBA after obsolescence has occurred. As a result, also for the whole simulation period with stationary demand in the beginning followed by sudden obsolescence later, the bias reduction for the TSB compared to Croston and SBA is much larger than for the case with stationary demand throughout the simulation period. For the same reason, TSB and SES have lower MSEs than Croston and SBA.

### 5.4. Summary of the results

Combining the results of Sections 5.1–5.3, it appears that TSB is the only method that performs well (compared to the other methods) in terms of both bias and variance, when all points in time are considered, for all the simulated scenarios with stationary demand, linearly decreasing demand and sudden obsolescence. However, we must remark that it is important to use the 'right' values for the smoothing constants of TSB. Too large values lead to poor performance under stationary demand, but too small values render the method slow in its response to changes in the demand, thereby deteriorating the performance under non-stationary demand. We will return to this issue in the next section.

## 6. Conclusions and extensions

A new method was proposed to forecast intermittent demand. This method exponentially smoothes the demand probability in every period, and the demand size after every positive demand. Contrary to the often applied Croston method and SBA modification, the new method is unbiased when considering an arbitrary point in time and can be used to deal with obsolescence issues

**Table 12**
ME and MSE results – sudden obsolescence ($c = 0.90$; $p = 0.5$).

| Estimator | ME | | | | MSE | | | |
|---|---|---|---|---|---|---|---|---|
| | α Smoothing constant | | | β | α Smoothing constant | | | β |
| | 0.10 | 0.20 | 0.30 | | 0.10 | 0.20 | 0.30 | |
| SBA | 1.0095 | 1.0205 | 0.9819 | | 9.9411 | 10.4586 | 10.8201 | |
| Croston | 1.1137 | 1.2416 | 1.3263 | | 10.2133 | 11.1690 | 12.1234 | |
| SES | 0.1677 | 0.0840 | 0.0560 | | 7.9357 | 8.2382 | 8.6583 | |
| TSB | 0.7951 | 0.8473 | 0.8672 | 0.01 | 8.9219 | 9.3488 | 9.7546 | 0.01 |
| | 0.6074 | 0.6473 | 0.6631 | 0.02 | 8.4298 | 8.7528 | 9.0679 | 0.02 |
| | 0.4769 | 0.5081 | 0.5208 | 0.03 | 8.1738 | 8.4408 | 8.7057 | 0.03 |
| | 0.3836 | 0.4086 | 0.4189 | 0.04 | 8.0286 | 8.2631 | 8.4986 | 0.04 |
| | 0.3152 | 0.3356 | 0.3444 | 0.05 | 7.9401 | 8.1544 | 8.3713 | 0.05 |
| | 0.1501 | 0.1599 | 0.1647 | 0.10 | 7.7895 | 7.9641 | 8.1429 | 0.10 |
| | 0.0606 | 0.0659 | 0.0696 | 0.20 | 7.8045 | 7.9639 | 8.1269 | 0.20 |
| | 0.0314 | 0.0362 | 0.0405 | 0.30 | 7.8952 | 8.0602 | 8.2319 | 0.30 |

by always providing up-to-date forecasts, also after long intervals with zero demand.

By using separate smoothing constants for the demand probability and demand size, the new method can be 'tuned' for demand processes with different (suspected) levels of non-stationarity. The variance expression that was derived under the assumption of stationarity is helpful in this respect, as the smoothing constants need to balance the variance against the speed of adaptation. A numerical investigation confirmed that TSB is suitable for situations with both stationary and non-stationary demand.

However, further research on what values to use for the smoothing constants is certainly needed. This should lead to recommendations on what values to use under which circumstances, especially related to the suspected levels of non-stationary demand behaviour. In addition, it may be possible to indicate ranges of values for which TSB generally performs well. Although the use of situation-dependent smoothing constants should certainly be advocated, some users might prefer to use fixed values for simplicity.

Another important avenue for further research is to empirically test the performance of TSB against that of other methods. This line of research could also address the above mentioned issue of how to set the smoothing constants. Furthermore, the issue of bias can be further explored by considering other bias measures, such as "periods in stock" as proposed by Wallström and Segerstedt (2010).

In further comparative studies, different smoothing parameters for the inter-demand interval and demand size for Croston and SBA could also be considered, as this could influence their comparative performance with TSB.

Finally, for both fast moving and intermittent demand, the link between forecasting and inventory control deserves more attention. There is an abundance of literature for both fields, but very few contributions that bridge the two.

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## Appendix A

Let us consider an intermittent, stationary demand process $p_t$ with demand probability $p$. So we have $E(p_t) = p$ and $Var(p_t) = p(1-p)$. Furthermore, we define the EWMA estimate as

$$p'_t = \sum_{i=0}^{\infty} \beta(1-\beta)^i p_{t-i}.$$

Then we have

$$E[p'_t] = \sum_{i=0}^{\infty} \beta(1-\beta)^i E[p_{t-i}] = \beta\, p \sum_{i=0}^{\infty} (1-\beta)^i = \frac{\beta}{1-(1-\beta)} p = p$$

and

$$\begin{aligned}
V[p'_t] &= E\left[\left\{\left(\sum_{i=0}^{\infty} \beta(1-\beta)^i p_{t-i}\right) - E[p'_t]\right\}^2\right] \\
&= E\left[\left\{\left(\sum_{i=0}^{\infty} \beta(1-\beta)^i p_{t-i}\right) - p\right\}^2\right] \\
&= E\left[\left\{\sum_{i=0}^{\infty} \beta(1-\beta)^i (p_{t-i} - p)\right\}^2\right] \\
&= \sum_{i=0}^{\infty} \left(\beta(1-\beta)^i\right)^2 E\left[(p_{t-i}-p)^2\right] = \sum_{i=0}^{\infty} \left(\beta(1-\beta)^i\right)^2 p(1-p) \\
&= p(1-p)\beta^2 \sum_{i=0}^{\infty} ((1-\beta)^2)^i = p(1-p)\beta^2 \frac{1}{1-(1-\beta)^2} \\
&= p(1-p)\frac{\beta}{2-\beta}.
\end{aligned}$$

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